

Smart Digital Product Recommendation Platform

Web Application User Guide and Demo Walkthrough

CP3407 Project Group PA-08

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Github Repository: <https://github.com/Chu-Junjie/CP3407-PROJECT>

Github Interactive web page: <https://chu-junjie.github.io/CP3407-PROJECT/>

Github README File: <https://github.com/Chu-Junjie/CP3407-PROJECT/blob/main/README.md>

The screenshot shows the homepage of the 'Smart Digital Product Recommendation' web application. At the top, there is a navigation bar with a blue 'S' logo and the text 'Smart Digital Product Recommendation'. Below this is a section titled 'Account & saved history' with a subtext: 'Search without an account, or log in to save and reopen searches.' This section contains input fields for 'Username (for registrati', 'Email', and 'Password (8+ character:' followed by 'Log in' and 'Register' buttons. The main section is titled 'Find My Perfect Device' and includes a description: 'Choose your requirements below. The page sends structured filters to the Flask recommendation API and renders products returned from the database-backed service.' Below this are several dropdown menus for filtering: 'Product type' (set to 'Laptop'), 'Maximum budget' (set to 'Under \$1,000'), 'Main use' (set to 'Everyday use'), 'Preferred brand' (set to 'Any brand'), and 'Brand to exclude' (set to 'Do not exclude a brand'). A large blue button labeled 'Find My Device' is positioned below these filters. At the bottom of the page, there is a small disclaimer: 'CP3407 Project - Recommendation results depend on the connected backend data. Product links may be manufacturer pages rather than checkout pages.'

Figure 1. Final homepage and recommendation input interface.

Guide purpose

This guide explains how a user accesses the web application, enters product needs, reviews database-backed recommendations, compares products, saves favorites, submits feedback and shares results. It is designed as a concise walkthrough for demonstration users.

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1. System overview

The Smart Digital Product Recommendation Platform is a web-based decision-support application for users who want to find suitable digital products without manually comparing many specifications. The final interface lets users submit structured product preferences, view ranked recommendations, compare products, save favorites, reopen search history, submit feedback and share recommendation states.

Area	Delivered user-facing function
Recommendation input	Users select product type, maximum budget, main use, preferred brand and excluded brand.
Database-backed results	The frontend sends structured filters to the Flask recommendation API and renders products returned by the backend.
Top recommendations	A Top Recommendations page shows ranked results with product names, prices, specs, score and reason.
Account center	Logged-in users can manage profile details, favorites and saved search history.
Comparison	Users can select up to three products and compare database-returned specification fields side by side.
Feedback and sharing	Users can vote Helpful / Not Helpful and generate a shareable recommendation state.

Reviewer note

The screenshots in this guide show the current user interface and the intended demonstration path. Product data comes from the connected backend and public/educational data sources rather than live retail inventory.

2. Access and user modes

The application supports both anonymous search and authenticated use. A visitor can search without logging in. A logged-in user can also save favorite products and reopen search history. This design keeps the basic recommendation flow simple while allowing persistent account features for users who want to continue their product comparison later.

Mode	What the user can do
Anonymous user	Search for recommendations, review product cards, compare selected products, view product/source links, submit feedback where enabled and share results.
Logged-in user	Use all anonymous functions, plus save favorite products, manage account state and reopen saved search history.

3. Homepage and search form

The homepage is the main entry point. The user can either search immediately or log in/register first. The form is intentionally structured so that users can express product requirements quickly without needing to know detailed hardware terminology.

Figure 2. Homepage with account controls and product recommendation filters.

Control	Supported options / behaviour
Account controls	Username, email and password fields support registration and login. Search remains possible without an account.
Product type	Laptop, Smartphone, Tablet, Headphones, Smart watch.
Maximum budget	Under \$500; Under \$1,000; Under \$1,500; Under \$2,000; Under \$3,000; Under \$5,000; No budget limit.
Main use	Everyday use; Study; Office and business; Gaming; Creative and video editing; Travel and portability; Photography and video; Long battery life; Fitness.
Preferred brand	Any brand; Apple; Samsung; HP; Sony; Other brands.
Brand to exclude	Do not exclude a brand; Apple; Samsung; HP; Sony; Other brands.
Find My Device	Submits the selected filters to the recommendation API and opens the recommendation results page.

Design rationale

The homepage uses dropdown controls to reduce invalid input, support repeatable testing and make the recommendation request easier to explain during demonstration.

4. Account, favorites and search history

The My Account page gives authenticated users a central place to manage profile state, favorite products and saved searches. Users can remove saved favorites, reopen previous searches, and select 2 or 3 favorite products from the same product type to compare them directly from the account page. This supports a realistic shopping workflow, where a user may want to return to previous recommendations, review saved products and compare shortlisted items later.

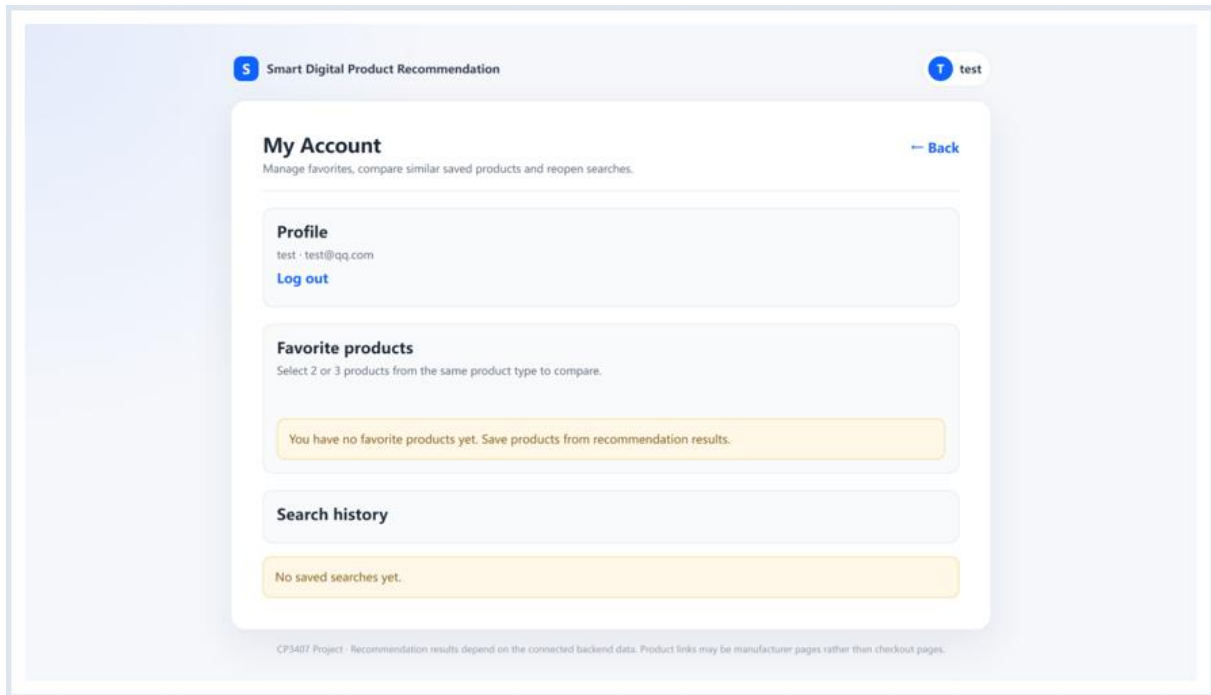


Figure 3. My Account page with profile, favorite products and search history sections.

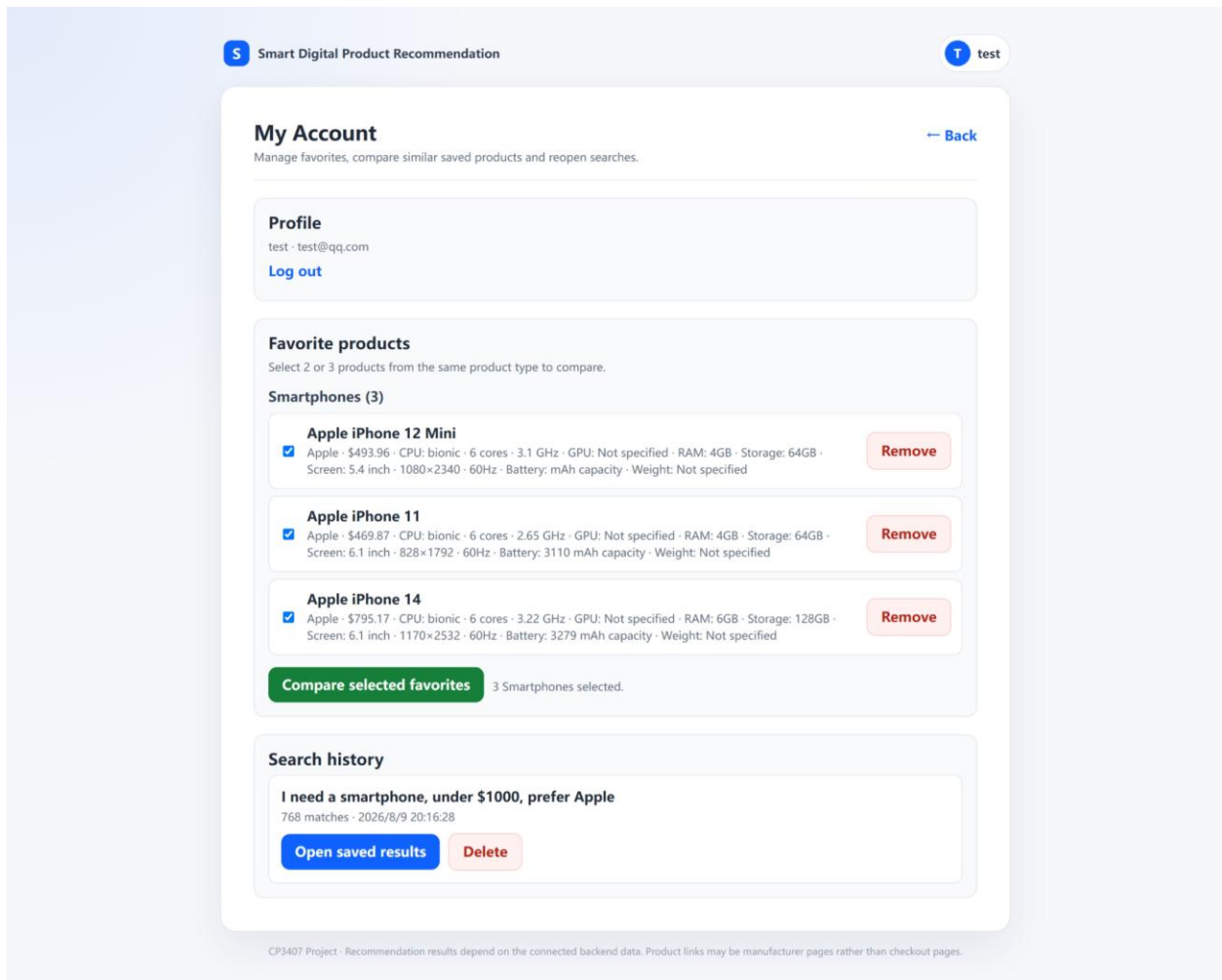


Figure 4. My Account page showing saved favorite products, compare-selected-favorites support and saved search history.

In this example, the user is logged in, has saved multiple smartphone favorites, and can compare up to three selected favorites directly from the account page. The same page also stores and reopens previous recommendation searches.

Section	Purpose
Profile	Shows the current signed-in demonstration account and provides logout access.
Favorite products	Lists products saved from recommendation results. The user can later compare 2 or 3 products from the same product type.
Search history	Stores previous searches when available so the user can reopen recommendation contexts.

5. Recommendation results page

After the user clicks Find My Device, the application displays ranked recommendation results. The page records the active request and shows products returned by the backend. Each result card includes product identity, price, category, relevant specifications, match score and action buttons.

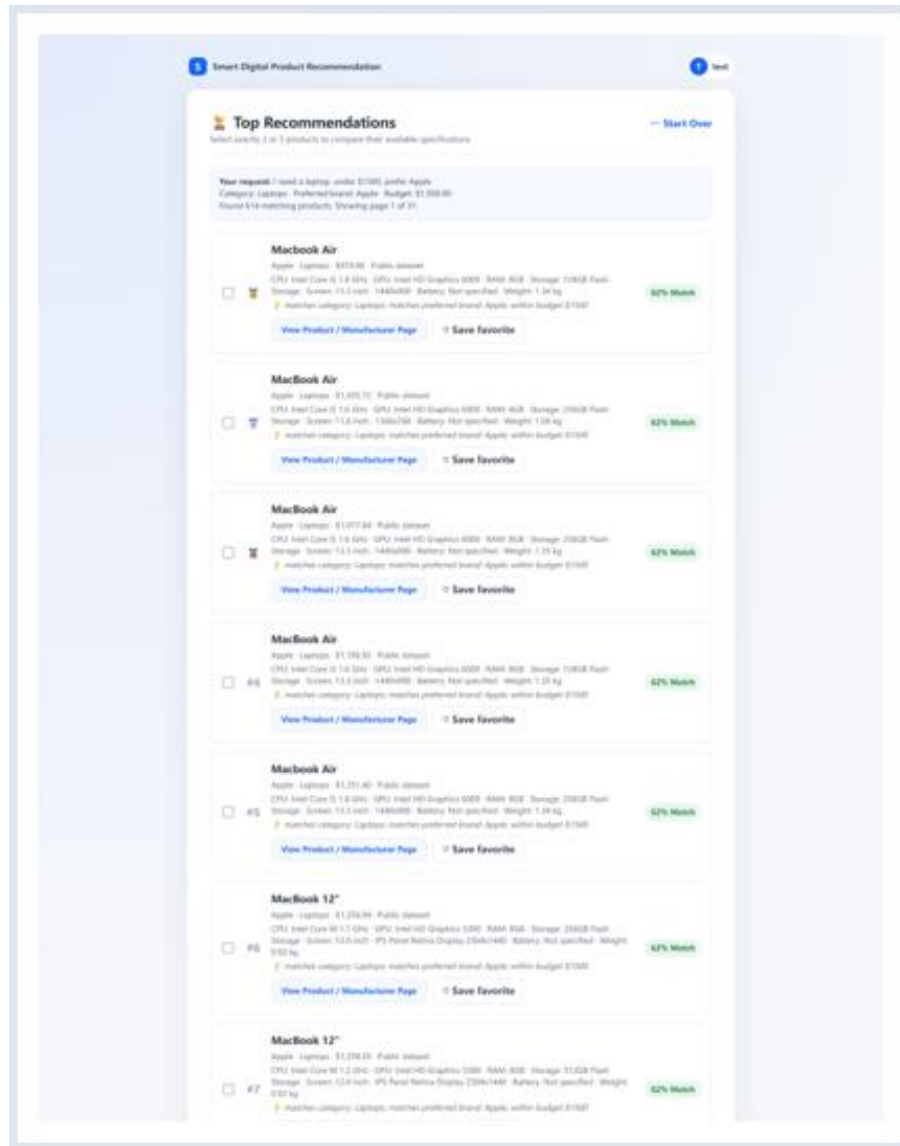


Figure 5. Top of the results page showing the request summary and ranked recommendation cards.

Element	User value
Request summary	Confirms the selected product type, main use, budget and excluded brand.
Product card	Displays product name, category, source type, technical fields, match score and reason.
Selection checkbox	Lets the user choose products for comparison.
View Product / Manufacturer Page	Opens a relevant product or manufacturer/source page when a safe link is available.
Save favorite	Saves the product to the authenticated user's favorite products.

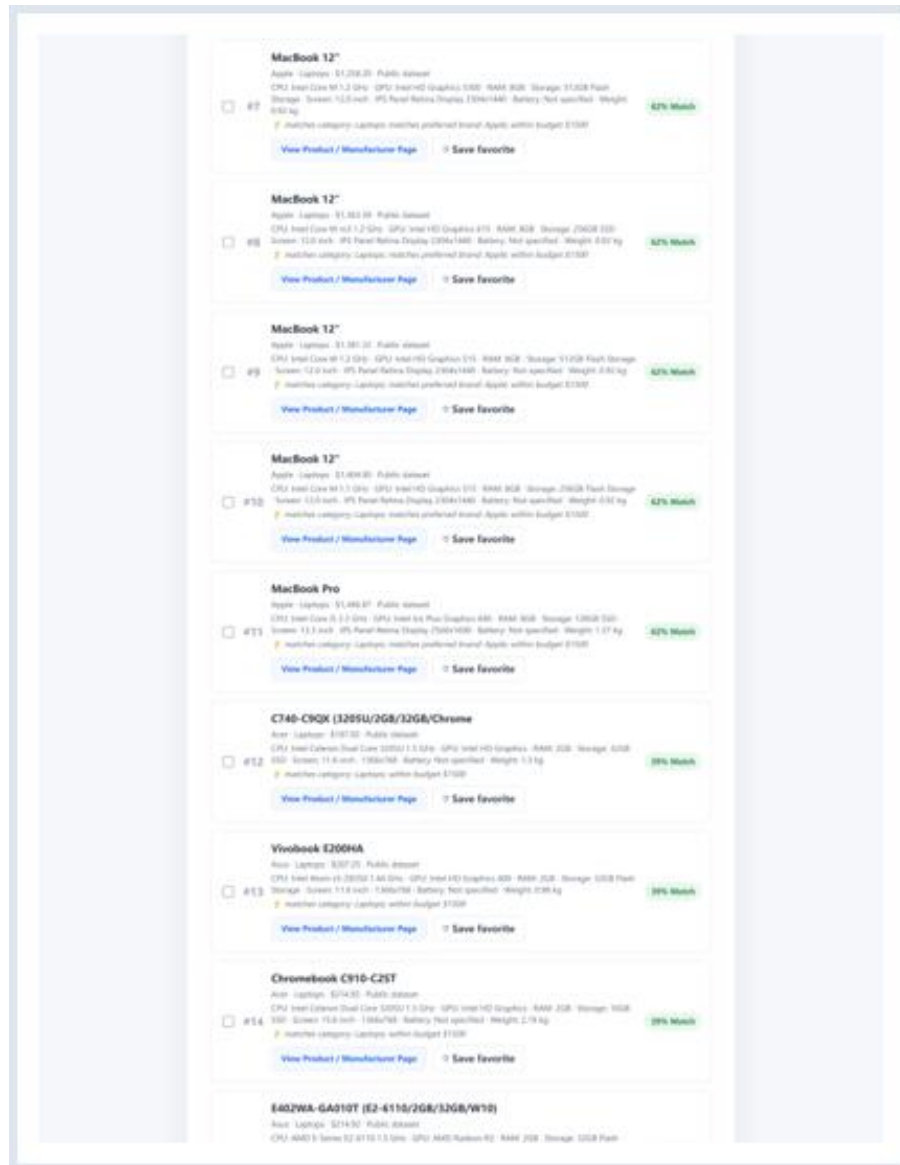


Figure 6. Middle section of recommendation results with additional products and repeated card actions.

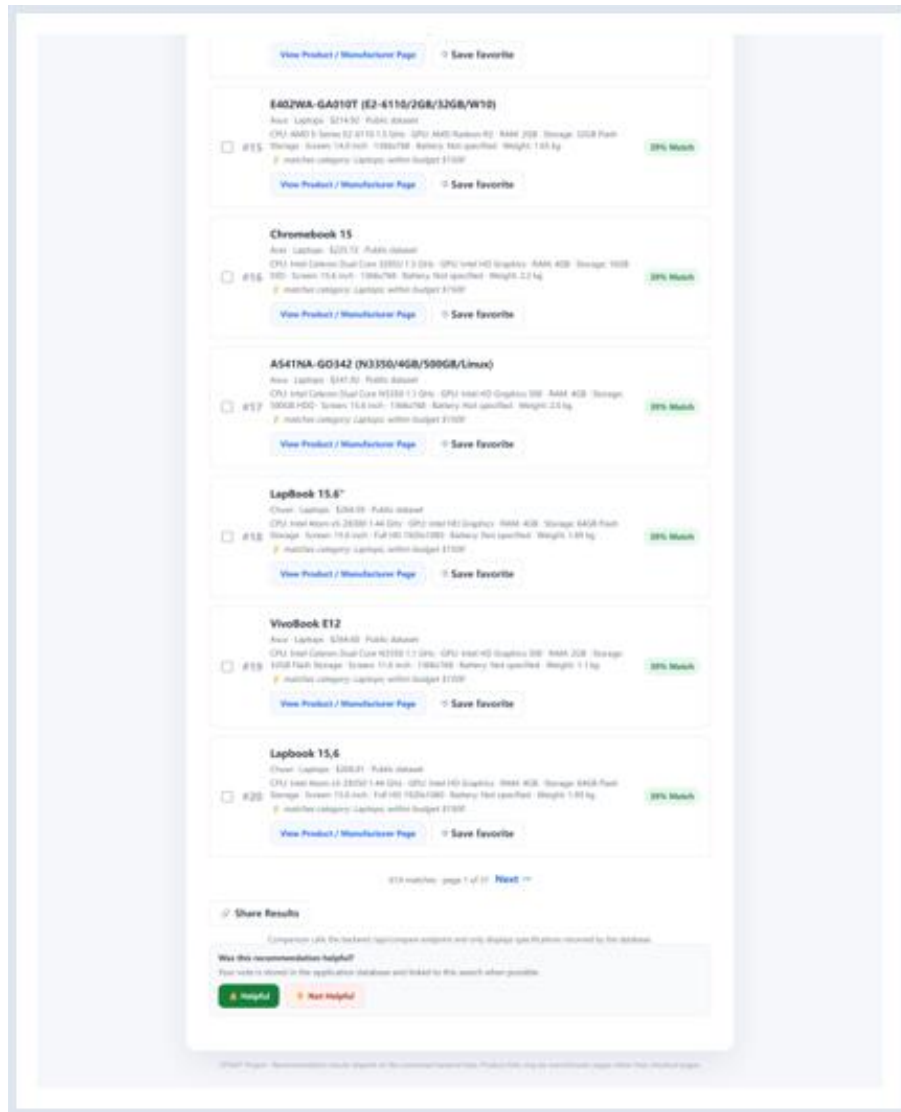


Figure 7. Bottom of the results page with pagination, share and feedback controls.

6. Product links, feedback and sharing

The results page includes actions that support decision-making and project feedback. Product/source links help users inspect external product information, while feedback and sharing support communication and improvement of the recommendation experience.

Action	How it works	Evidence / note
View Product / Manufacturer Page	Opens the product, source or manufacturer page associated with the selected result.	The guide does not claim guaranteed checkout availability.
Save favorite	Stores the selected product under the logged-in user's favorites.	Useful for later account-based comparison.
Share Results	Generates/copies the current recommendation page state or URL for reuse.	The share state should not expose passwords, tokens or private account data.
Helpful / Not Helpful	Submits a user vote on the recommendation result.	The vote is stored in the application database and linked to the search when possible.

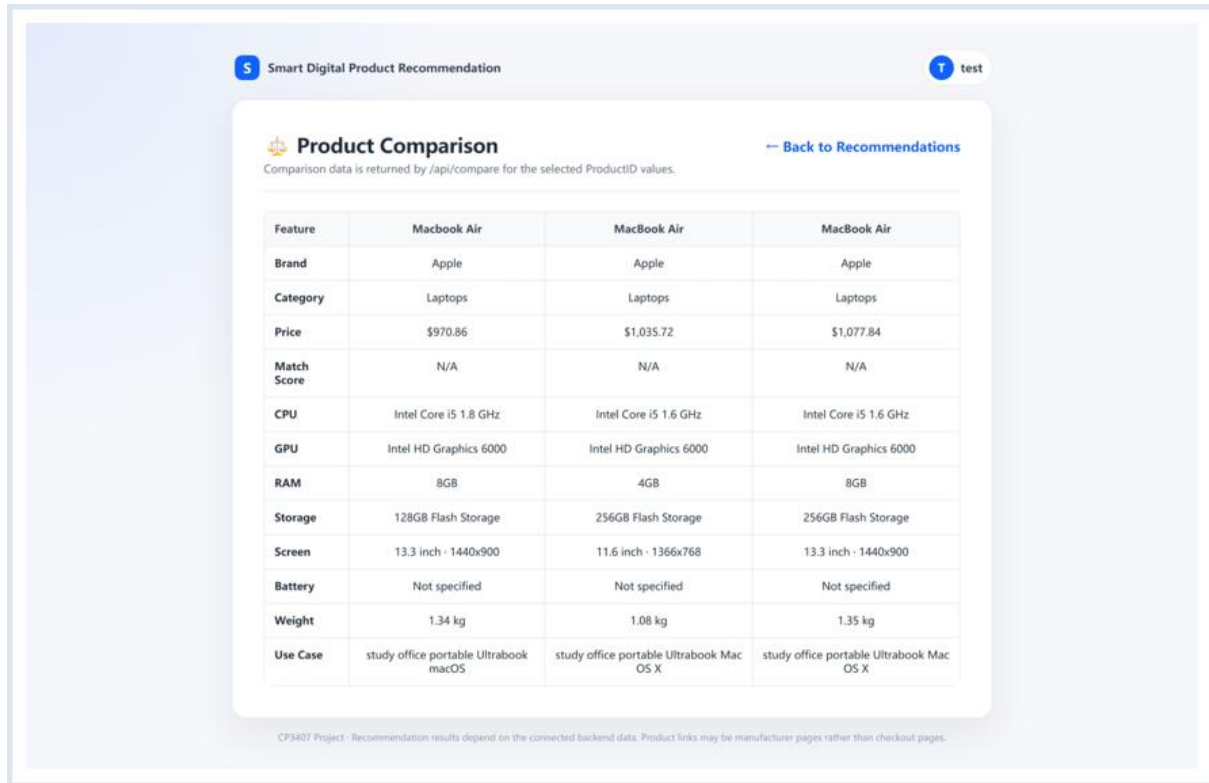
Feedback wording shown in the interface

The application informs the user: "Was this recommendation helpful? Your vote is stored in the application database and linked

to this search when possible."

7. Product comparison

The product comparison page supports side-by-side review of up to three selected products. The comparison table is returned by `/api/compare` for the selected ProductID values. This keeps the frontend consistent with database-backed product and specification fields.



The screenshot shows a web application interface for "Smart Digital Product Recommendation". The main content area is titled "Product Comparison" and includes a "Back to Recommendations" link. Below the title, a note states: "Comparison data is returned by /api/compare for the selected ProductID values." A table displays the comparison of three MacBook Air models. The table has four columns: Feature, MacBook Air, MacBook Air, and MacBook Air. The rows list various features including Brand, Category, Price, Match Score, CPU, GPU, RAM, Storage, Screen, Battery, Weight, and Use Case. The data is as follows:

Feature	Macbook Air	MacBook Air	MacBook Air
Brand	Apple	Apple	Apple
Category	Laptops	Laptops	Laptops
Price	\$970.86	\$1,035.72	\$1,077.84
Match Score	N/A	N/A	N/A
CPU	Intel Core i5 1.8 GHz	Intel Core i5 1.6 GHz	Intel Core i5 1.6 GHz
GPU	Intel HD Graphics 6000	Intel HD Graphics 6000	Intel HD Graphics 6000
RAM	8GB	4GB	8GB
Storage	128GB Flash Storage	256GB Flash Storage	256GB Flash Storage
Screen	13.3 inch · 1440x900	11.6 inch · 1366x768	13.3 inch · 1440x900
Battery	Not specified	Not specified	Not specified
Weight	1.34 kg	1.08 kg	1.35 kg
Use Case	study office portable Ultrabook macOS	study office portable Ultrabook Mac OS X	study office portable Ultrabook Mac OS X

At the bottom of the page, a small disclaimer reads: "CP3407 Project - Recommendation results depend on the connected backend data. Product links may be manufacturer pages rather than checkout pages."

Figure 8. Product comparison page showing three selected products side by side.

Comparison rule	Explanation
Maximum selection	The user can compare up to 3 products.
Recommended selection	2 or 3 products are easiest to read and compare.
Data source	Comparison fields are returned by the backend API for selected ProductID values.
Displayed fields	Brand, category, price, match score, CPU, GPU, RAM, storage, screen, battery, weight and use case where available.
Missing values	Missing fields are shown honestly as N/A or Not specified rather than being invented.

8. Data boundary and limitations

The application is an educational recommendation prototype. The interface and documentation should describe the data honestly so that users understand what the system does and does not claim.

Boundary	Explanation
Product catalogue	Uses connected backend data and public/educational product datasets rather than a live market-wide inventory.
Prices	Prices are historical or dataset-derived values and should not be presented as live retail prices.
Product links	Links may lead to product, manufacturer or source pages and are not guaranteed checkout links.

Specifications

The application displays backend-provided fields; missing fields are shown as N/A / Not specified.

9. Demonstration checklist

The following checklist can be used by a marker, reviewer or team member during a live demonstration.

Step	Demonstration action
1	Open the homepage and confirm the account/search controls are visible.
2	Search anonymously using product type, budget, main use and brand controls.
3	Confirm recommendation cards are returned and include product details and match scores.
4	Open a View Product / Manufacturer Page link.
5	Select 2 or 3 products and open the comparison table.
6	Register or log in with a demonstration account.
7	Save a favorite product and confirm it appears in My Account.
8	Submit Helpful or Not Helpful feedback.
9	Use Share Results and reopen the state in another browser/private session.
10	Check mobile layout and keyboard navigation for the primary flow.

Recommended evidence files

For final review, keep screenshots of the homepage, account page, recommendation page, comparison page, browser Network evidence for API calls, and any deployed smoke-test results.